

■ Music Instrument from Waste

Detailed Lesson Plan for Early Childhood Educators

Subject	Arts & Science — Sound and Music	Grade Level	Early Childhood / KG – Class 2
Duration	60 – 90 minutes	Class Size	15 – 30 students
Setting	Classroom / Activity Hall	Language	English (with Hindi/regional cues)

■ Learning Objectives

By the end of this lesson, students will be able to:

- Explain that sound is produced by vibration (things shaking fast).
- Identify how different materials produce different sounds.
- Classify simple instruments into Shakers, Tappers, and Blowers.
- Recognise common Indian musical instruments (tabla, dholak, bansuri, ghungroo, manjira).
- Create a functional bottle shaker from waste materials.
- Articulate the importance of reusing waste to protect the environment.

■ Materials Required

For each student/pair:

- 1 empty, clean plastic bottle (with tight cap)
- A handful of rice, dal, or small pebbles (filling)
- Stickers, paint pens, or ribbon for decoration

For demonstration & exploration:

- 1 filled bottle and 1 empty bottle (for sound demo)
- Steel dabbas / tiffin boxes, rubber bands, cardboard tubes, bottle caps
- A speaker / phone to play 'Twinkle Twinkle Little Star' and 'Johny Johny'

Optional enrichment:

- Pictures or video clips of tabla, dholak, bansuri, shehnai, ghungroo, manjira

■ Lesson Flow (Step-by-Step)

Phase 1 Warm-Up — Body Percussion Orchestra ■ 10 min	
Hook	Ask: 'Did you know your own body is a musical instrument?' Demonstrate clapping, knee-patting, foot-stomping, and tongue-clicking. Invite children to join.
Connect	Ask: 'Has anyone ever tapped a cardboard box at home and turned it into a drum?' Celebrate responses with 'Wah!'
Transition	Tell students: 'Today we will make real instruments — from things we usually throw away!'
Phase 2 Story Time — Riya and the Noisy Kitchen ■ 10 min	
Narrate	Tell the story of Riya who walks into the kitchen and hears tap-tap (spoon on thali), clang-clang (lid on pot), and dhum-dhum (hand on dabba).
Pose the Question	Ask: 'Why do things make sound when we hit them?' Accept all answers warmly.
Reveal the Concept	Explain: 'Sound happens when something SHAKES FAST — that shaking is called VIBRATION!' Have children say 'VIBRATION' together.
Phase 3 Sound Demo — Empty vs Filled Bottle ■ 8 min	
Demo	Shake an empty bottle → listen. Then shake a bottle filled with rice → compare.
Ask	'Which is louder? Why?' Guide students to see that more material = more vibration = louder sound.
Reinforce	Say: 'The rice or dal inside shakes and vibrates more, so we hear more sound!'
Phase 4 Sound Hunt — Exploring Materials ■ 10 min	
Activity	Place materials (plastic bottles, steel dabbas, rubber bands, cardboard tubes, bottle caps, pebbles, newspaper) on tables or in a centre.
Task	Each child/pair picks any 2 items. They tap, shake, or blow. They describe the sound to the class.
Discussion Prompts	'Is it loud or soft? High or low? Like which animal or which festival sound?'
Phase 5 Make Your Own Shaker! ■ 15 min	
Step 1	Take an empty, clean plastic bottle.
Step 2	Put a handful of rice, dal, or pebbles inside.

Step 3	Close the cap tightly (adult assists if needed).
Step 4	Decorate with stickers, paint pens, or ribbon — make it your own!
Safety Note	Ensure caps are screwed on firmly. No sharp edges. Adult supervises filling.

Phase 6 Shake Along to the Music! ■ 8 min	
Play Music	Play 'Twinkle Twinkle Little Star' — children shake their instruments to the beat.
Play Music 2	Play 'Johnny Johnny Yes Papa' — children follow the rhythm together.
Celebrate	Praise with: 'Very Good! Well Done! Wah!'

Phase 7 Indian Instruments Around Us ■ 7 min	
Blowers	Bansuri & Shehnai — blow air to make sound. Demo blowing through a cardboard tube.
Shakers	Ghungroo & Manjira — shake to make a jingling sound (like the bottle shaker!).
Tappers	Tabla & Dholak — tap to make drum beats (like the steel dabba).
Connection	'Our shaker is just like a ghungroo! Our dabba is just like a dholak!'

Phase 8 Key Concept Recap ■ 5 min	
Point 1	Sound is made by shaking or hitting things — that's VIBRATION.
Point 2	Different materials make different sounds — try it at home!
Point 3	Everyday things can become musical instruments!
Caring for Earth	Talk about reusing waste: 'Kabadiwala collects old things. Dadi uses biscuit tins as drums. We used a bottle! No waste — waste se banao!'

■ Assessment

Informal Observation

- Can the child describe the sound of each material (loud/soft, high/low)?
- Does the child correctly follow the 3-step shaker-making process?
- Can the child keep a basic rhythm while shaking along to the music?

Verbal Check (Quick questions at recap):

- 'What is vibration?' (Expected: things shaking fast)
- 'Name one instrument that is a shaker / tapper / blower.'
- 'Why did we use a waste bottle instead of throwing it away?'

Take-Home Indicator:

- Child brings the decorated shaker home and demonstrates it to family — a conversation starter about sound and reuse.

■ Differentiation

Support (Struggling Learners)	Extension (Advanced Learners)
• Pre-fill the bottle; child closes cap and decorates. • Provide picture cues for steps. • Pair with a buddy.	• Try 2–3 fillings (rice vs pebbles vs sand) and compare sounds. • Create a simple 4-beat pattern and teach it to a friend. • Draw and label their instrument.

■ Teacher Tips

- **Energy management:** Alternate active (body percussion, shaking) with calm (story, recap) phases to maintain focus.
- **Cultural hooks:** Use familiar words — dabba, thali, kabadiwala, bansuri — to make concepts relatable.
- **Praise language:** Use 'Wah!', 'Shabash!', 'Very Good!' to keep energy high and inclusive.
- **Noise management:** Agree on a 'freeze' signal (raised hand or a single clap) before the shaker activity begins.
- **Safety:** Check all caps are tightly closed before the shake-along. Supervise younger children during filling.
- **Time flex:** Phases 4 and 5 can overlap if materials are pre-set on tables. Phase 7 can be shortened or sent as a homework card.

■ Curriculum Links

- **Science / EVS:** Sound and vibration; properties of materials
- **Arts & Craft:** Creating an instrument; decorating with design
- **Music:** Rhythm, beat, Indian classical instruments
- **Environmental Education:** Reuse, reduce waste, care for the Earth
- **Language:** Describing sounds (loud/soft, high/low, jingling/thumping); vocabulary — vibration, rhythm, shaker, tapper, blower
- **Social Studies:** Music in Indian festivals, homes, and temples; cultural instruments

■ Extension Activities

Class Junk Orchestra (next session):

Assign each child one type: Shakers, Tappers, or Blowers. Collect a variety of waste items and build a class orchestra. Play a familiar nursery rhyme together. Record and play back!

Take-Home Challenge:

Ask families: 'Find one item at home that makes a sound. What type is it — shaker, tapper, or blower? Draw it and bring your drawing to school!'